CentraleSupélec

Order–disorder transition of β-brass (CuZn) by Metropolis Monte Carlo on a rigid lattice

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Abstract

We wrote a Metropolis Monte Carlo code that exchanges Cu and Zn atoms on a fixed body-centred cubic lattice of 432 sites, with a pair energy model limited to first and second neighbours. The code reproduces the exact values of the perfect B2 structure (cohesive energy 2.484 eV/atom) and of a random alloy. Scanning the temperature from 200 K to 1600 K, we follow the energy, the heat capacity, the long-range order parameter η, its susceptibility and the Cowley short-range order parameters. The three estimates of the transition temperature all give $T_c \approx 720$ K, close to the experimental value of about 740 K. Short-range order survives above T_c.

1 Introduction

β-brass is the CuZn alloy with about 50 % Zn. At low temperature it is ordered in the B2 (CsCl-type) structure: the atoms sit on a body-centred cubic (bcc) lattice, Cu on the cube corners and Zn on the cube centres. When the temperature rises, Cu and Zn atoms swap places more and more often. Around 740 K the distinction between the two kinds of sites disappears: the alloy becomes a disordered bcc solid solution [1–3]. This is a classic order–disorder transition, first described by Bragg and Williams with a mean-field theory [4].

In this project [5] the atoms do not move away from the lattice sites ("rigid lattice"): only the occupation of each site by Cu or Zn changes. The energy is a sum of pair interactions between first and second neighbours. We sample the equilibrium configurations with the Metropolis algorithm [6] and measure how the order is lost when the temperature increases.

2 Model and method

2.1 Lattice and neighbours

At low temperature CuZn has the ordered B2 structure (CsCl type), made of two interpenetrating simple cubic sublattices α and β. The cubic cell of parameter a contains two sites (Figure 1): the α site at the corner $(0, 0, 0)$, occupied by Cu in B2, and the β site at the centre $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, occupied by Zn. Whatever the species on them, the sites form a bcc lattice, which stays fixed during the simulation. The simulation box contains $6 \times 6 \times 6$ cells, that is $N = 2 \times 6^3 = 432$ sites (216 Cu and 216 Zn, $c_{\mathrm{Zn}} = 1/2$), with periodic boundary conditions in the three directions. The reference configuration at $T = 0$ K is the B2 cell repeated $6 \times 6 \times 6$ times.

Each site has two neighbour shells (Table 1): 8 first neighbours at distance $\sqrt{3}a/2$, all on the other sublattice, and 6 second neighbours at distance a, on the same sublattice. The list of these 14 neighbours is built once for every site at the start of the program, applying the periodic boundary

conditions (modulo 6 on each cell coordinate). This table never changes: only the occupation of the sites does.

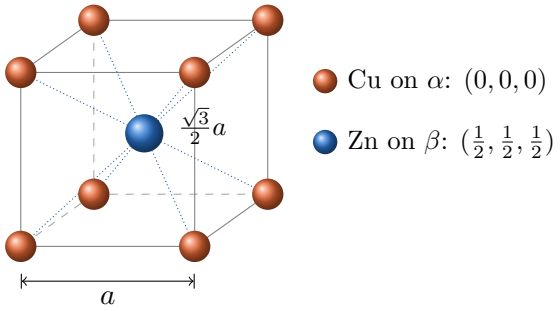


Figure 1: Cell of the B2 structure of CuZn (perfect order, $T = 0\,\mathrm{K}$). Cu atoms (orange) occupy the corners (sublattice α), Zn atoms (blue) the centre (sublattice β). The dotted lines are the 8 first-neighbour bonds of the centre atom (length $\sqrt{3}a/2$); second neighbours are along the cube edges (length a). In the disordered phase, Cu and Zn occupy both sublattices at random, and the lattice is simply bcc.

2.2 Energy

The energy of a configuration is

$$E = \sum_{\langle i,j\rangle_1} J_{\sigma_i\sigma_j} + \sum_{\langle i,j\rangle_2} I_{\sigma_i\sigma_j}, \tag{1}$$

where $\sigma_i \in \{\mathrm{Cu}, \mathrm{Zn}\}$ is the atom on site i, and the sums run over first- and second-neighbour pairs, each pair counted once. The pair energies are given in Table 1.

Table 1: Neighbour shells and pair energies of the model [5], in eV. J acts between first neighbours, I between second neighbours. The last column is the "ordering energy" $w = X_{\mathrm{CuCu}} + X_{\mathrm{ZnZn}} - 2X_{\mathrm{CuZn}}$ of each shell ($X = J$ or I): $w > 0$ favours unlike pairs, $w < 0$ favours like pairs.

Shell	Distance	Neighbours	Cu–Cu	Zn–Zn	Cu–Zn	w
1st neighbours (J)	$\sqrt{3}a/2$	8	-0.873	-0.338	-0.621	$+0.031$
2nd neighbours (I)	a	6	0.000	0.000	$+0.005$	-0.010

The first shell favours Cu–Zn pairs ($w_1 > 0$) and the second shell favours pairs of the same species ($w_2 < 0$). In the B2 structure the 8 first neighbours of every atom are of the other species and the 6 second neighbours of the same species, so both shells push towards B2 order.

2.3 Monte Carlo simulation

Principle. At temperature T, a configuration of energy E has the Boltzmann probability $P \propto e^{-E/k_BT}$. The mean value of a quantity would be a sum over all the configurations, but there are about 4×10^{128} ways to place 216 Cu atoms on 432 sites, so this sum is out of reach. A Monte Carlo simulation replaces it by an average over a sample of configurations drawn at random with the right probabilities. The Metropolis algorithm [6] builds this sample as a chain: from the current configuration it proposes a small random change, and accepts it with the probability

$$P_{\mathrm{acc}} = \min\left(1, e^{-\Delta E/k_BT}\right), \tag{2}$$

where ΔE is the energy change of the proposed move. A move that lowers the energy is always accepted; a move that raises it is accepted less and less often as $\Delta E/k_BT$ grows. This rule satisfies detailed balance, so after a transient the chain visits the configurations with their Boltzmann probabilities, and averages along the chain are thermal averages [7].

Steps of a run. The composition is fixed, so a move must exchange two atoms rather than change the species of one site (Kawasaki dynamics [8]). Each temperature is an independent run, made of the following steps:

1. *Initialisation.* Start from the perfect B2 structure and compute its energy E with Equation (1).
2. *Proposed move.* Draw a site i at random among the 432 sites, then one of its 8 first neighbours j at random.
3. *Energy change.* If i and j hold the same species, the exchange changes nothing: go to step 5. Otherwise compute ΔE. It only involves the bonds of i and j with their other neighbours (7 first and 6 second neighbours each), because the i–j bond itself stays a Cu–Zn bond. This local calculation has 26 bonds instead of the $432 \times 14/2 = 3024$ bonds of Equation (1).
4. *Metropolis test.* Accept the exchange with the probability of Equation (2): if $\Delta E \leq 0$ it is accepted; otherwise draw a uniform random number u between 0 and 1, and accept if $u \leq e^{-\Delta E/k_B T}$. If it is accepted, exchange the two atoms and update the energy, $E \leftarrow E + \Delta E$; if not, keep the configuration as it is.
5. *Repetition.* Go back to step 2. Each attempted exchange counts as one Monte Carlo step, accepted or not. A *sweep* is $N = 432$ steps, so that every atom is chosen once on average.
6. *Equilibration.* Do not measure anything during the first N_{eq} steps, while the system moves away from the B2 start (Section 3).
7. *Measurement.* Then continue for M sweeps and record the quantities of Section 2.4 once per sweep.

2.4 Measured quantities

After equilibration, we record once per sweep the energy E, the long-range order parameter and the two Cowley parameters:

$$\eta = \frac{N_{\mathrm{Cu}}^{(\alpha)} - N_{\mathrm{Cu}}^{(\beta)}}{N_{\mathrm{Cu}}^{(\alpha)} + N_{\mathrm{Cu}}^{(\beta)}}, \qquad \alpha_n = 1 - \frac{p_n^{\mathrm{Zn|Cu}}}{c_{\mathrm{Zn}}} \quad (n = 1, 2), \tag{3}$$

where $N_{\mathrm{Cu}}^{(\alpha)}$ is the number of Cu atoms on sublattice α, $p_n^{\mathrm{Zn|Cu}}$ the probability that a neighbour of shell n of a Cu atom is a Zn atom, and $c_{\mathrm{Zn}} = 1/2$ [9]. The perfect B2 order gives $\eta = 1$, $\alpha_1 = -1$ and $\alpha_2 = +1$; a random alloy gives $\eta = \alpha_1 = \alpha_2 = 0$. In a finite box the ordered state can also form with Cu on β ($\eta < 0$), and near T_{c} the system goes back and forth between the two. We therefore average $|\eta|$, which measures the amount of order whatever its sign.

From the series of M measurements we compute the means $\langle E \rangle$, $\langle |\eta| \rangle$, $\langle \alpha_n \rangle$, and two response functions from the fluctuations [7]:

$$C_v = \frac{\langle E^2 \rangle - \langle E \rangle^2}{k_B T^2}, \qquad \chi = \frac{N \left(\langle \eta^2 \rangle - \langle |\eta| \rangle^2 \right)}{k_B T}. \tag{4}$$

C_v is the heat capacity of the whole box (in eV/K) and χ the susceptibility of the order parameter (in eV^{-1}).

2.5 Simulation conditions

Table 2 gathers the conditions of all the calculations of this report.

Table 2: Conditions of all the calculations of this report. The numbers of steps refer to attempted exchanges; 1 sweep = 432 steps.

Lattice	bcc, $6 \times 6 \times 6$ cubic cells, $N = 432$ sites, periodic boundary conditions
Composition	216 Cu + 216 Zn ($c_{\mathrm{Zn}} = 1/2$), conserved by the moves
Energy	pair model of Equation (1), parameters of Table 1; $k_B = 8.617\,333\,262 \times 10^{-5}$ eV/K
Moves	exchange of a random site with a random first neighbour, Metropolis acceptance
Starting configuration	perfect B2 (Cu on α, Zn on β) at every temperature
Equilibration test	200 000 steps (463 sweeps) from B2 at each temperature, energy recorded after every step
Equilibration	$N_{\mathrm{eq}} = 50\,000$ steps (116 sweeps), not measured
Measurement	$M = 4000$ sweeps, one measurement of E, η, α_1, α_2 per sweep
Temperatures (K)	200, 300, 400, 500, 550, 600, 650, 675, 700, 710, 720, 730, 740, 750, 760, 775, 800, 850, 900, 1000, 1200, 1600

3 Equilibration

Every run starts from the perfect B2 structure, which is not representative of a finite temperature. The first steps must be discarded until the system has "forgotten" this initial state. To choose this number of steps, we record the energy after every step during 200 000 steps at each temperature of the scan (Figure 2). The equilibrium energy E_{eq} and the size σ of its fluctuations are the mean and standard deviation over the second half of the run; equilibrium is reached at the first step where $E \geq E_{\mathrm{eq}} - \sigma$.

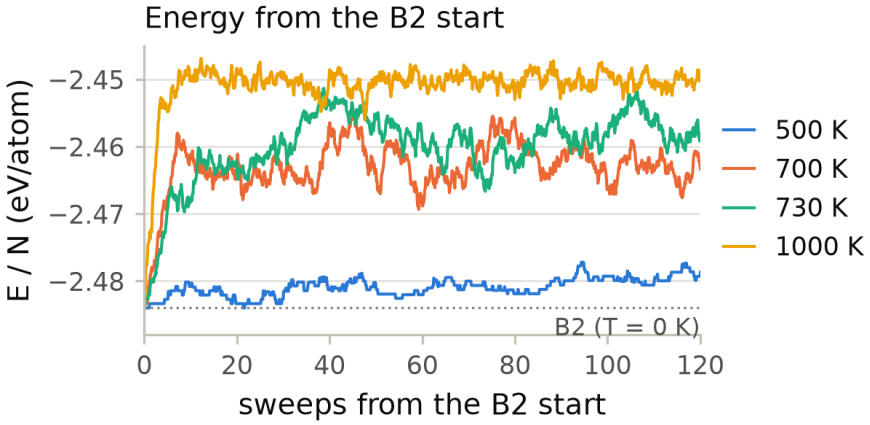


Figure 2: Equilibration from the perfect B2 structure: energy per atom during the first 120 sweeps at four temperatures (one value every 50 steps). The dotted line is the B2 energy.

At low temperature (200–300 K) the B2 structure is already the equilibrium state and almost no exchange is accepted. At high temperature, disorder sets in within a few thousand steps. The slowest relaxations are found below and near the transition, with at most 11 806 steps (27 sweeps) at 730 K. This estimate is itself noisy, because the energy fluctuates strongly near T_{c}. We therefore discard $N_{\mathrm{eq}} = 50\,000$ steps at every temperature, more than four times the slowest relaxation.

4 Physical quantities and transition temperature

At each temperature of Table 2, we ran the Metropolis algorithm from the B2 structure until equilibrium (N_{eq} steps, Section 3). Once equilibrium was reached, we continued the run for $M = 4000$ sweeps and calculated the physical quantities of Section 2.4 from the recorded series. All the values are listed in Table 4.

4.1 Evolution with temperature

Figure 3 shows what the configurations look like. At 300 K the layer is a perfect checkerboard of Cu and Zn. Near T_c the order is only partial, and at 1200 K the atoms are spread at random over the two sublattices.

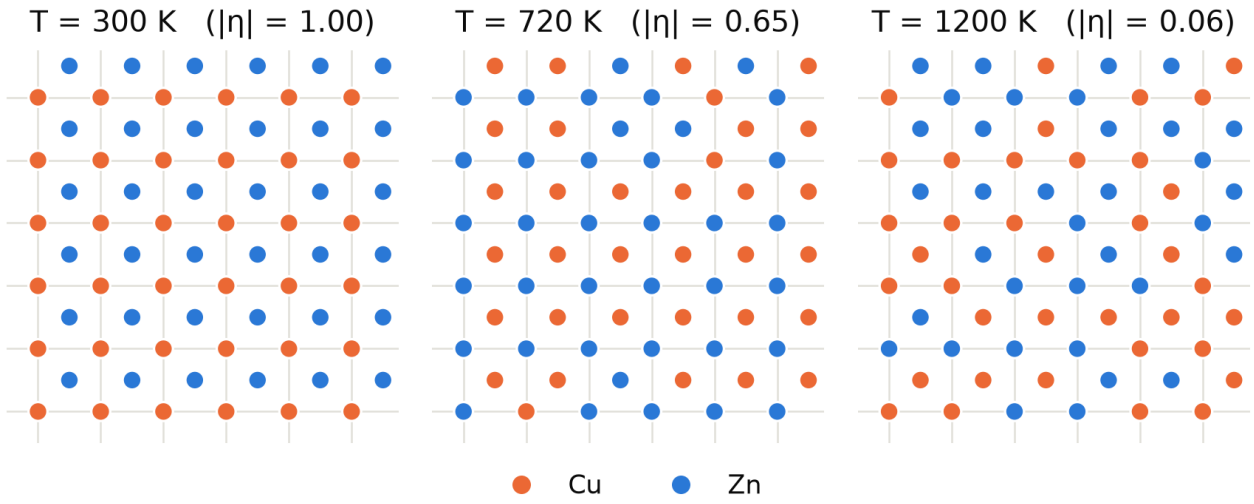


Figure 3: Final configuration of the runs at three temperatures, seen from above: one layer of the box, with the α sites of height $z = 0$ at the grid nodes and the β sites of height $z = a/2$ at the centres of the squares (6×6 cells, 72 of the 432 atoms). $|\eta|$ is computed on the whole box.

Energy and heat capacity. The mean energy (Figure 4, left) starts from the B2 value and rises steeply between 600 and 800 K, where most of the disorder appears. At high temperature it tends towards the energy of the ideal random alloy, $E_{\mathrm{dis}}/N = 4\bar{J} + 3\bar{I} = -2.4455$ eV/atom, where $\bar{J}$ and $\bar{I}$ are the averages of the pair energies weighted by the probabilities $\frac{1}{4}, \frac{1}{4}, \frac{1}{2}$ of Cu–Cu, Zn–Zn and Cu–Zn pairs. Even at 1600 K the energy remains 0.0023 eV/atom below this value, because of the short-range order discussed below. The heat capacity (Figure 4, right) has a clear peak at 720 K; its value at 700 K is almost the same, so the top of the peak is between 700 and 720 K.

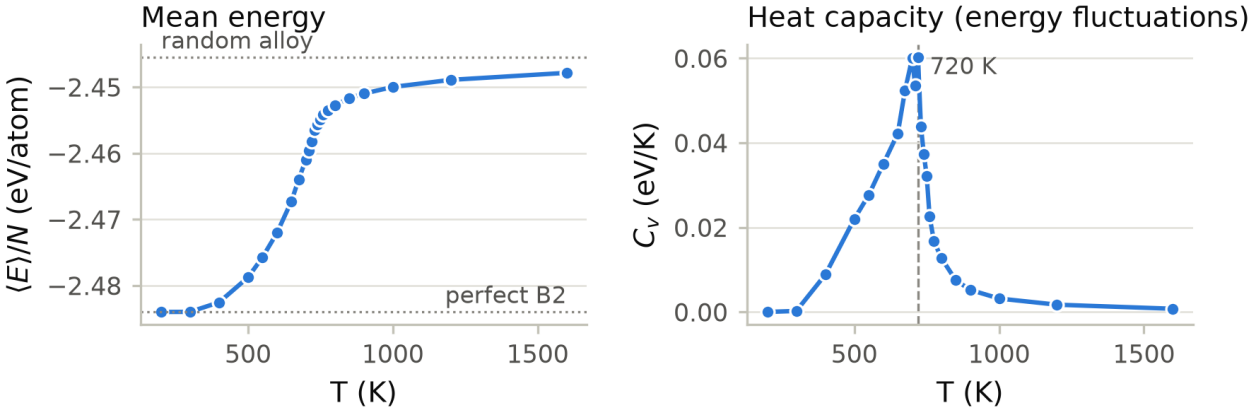


Figure 4: Left: mean energy per atom versus temperature; the dotted lines are the perfect B2 energy (−2.484 eV/atom) and the energy of an ideal random alloy (−2.4455 eV/atom). Right: heat capacity C_v of the box (432 atoms), from the energy fluctuations (Equation (4)); the dashed line marks its maximum.

Long-range order. $\langle|\eta|\rangle$ (Figure 5, left) decreases slowly up to about 600 K, then drops fast between 650 and 800 K, and finally approaches the value of a random lattice, about 0.04. This value

is not zero because, with only 216 Cu atoms, chance alone puts a few more of them on one sublattice. The susceptibility (Figure 5, right) has a sharp peak at 720 K, almost equal to its value at 730 K.

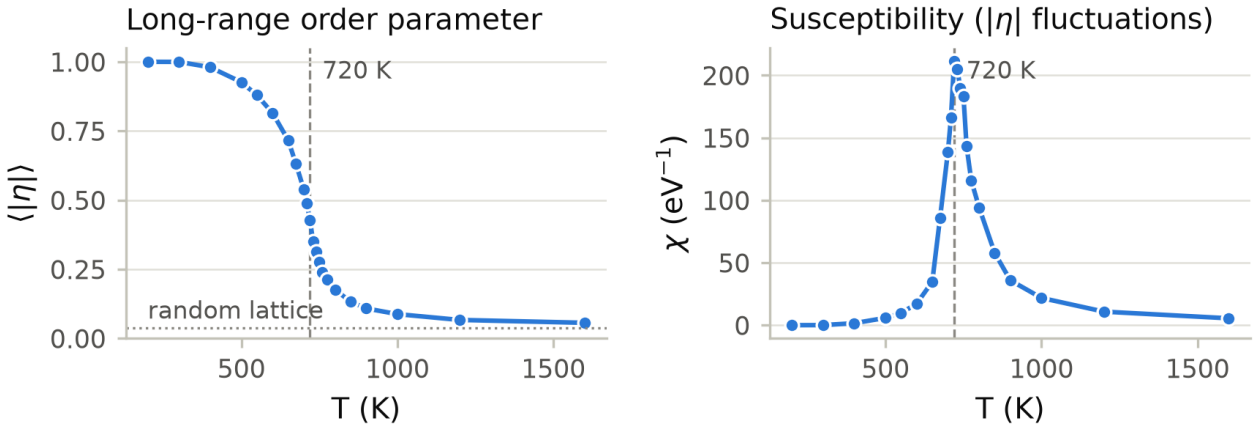


Figure 5: Left: mean absolute long-range order parameter; the dotted line is the value for random lattices (finite-size floor), the dashed line marks the steepest drop. Right: susceptibility of the order parameter from the fluctuations of $|\eta|$ (Equation (4)); the dashed line marks its maximum.

Figure 6 shows the distribution of $|\eta|$. Below T_c it is narrow around a large value; above T_c it is narrow close to zero. Near T_c it is very broad: the system wanders between ordered and disordered states.

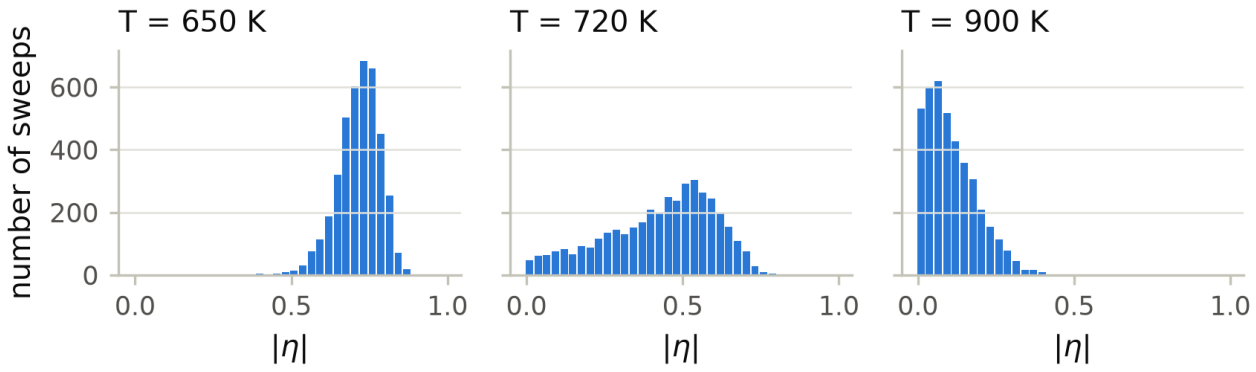


Figure 6: Histograms of the 4000 values of $|\eta|$ measured (one per sweep) below, near and above the transition. In our box $|\eta|$ can only take the values $2k/216$; each bin contains three of them (width $6/216 \approx 0.028$).

Short-range order. Figure 7 shows the Cowley parameters. If the only order were the long-range order, a Cu atom on α would find a Zn neighbour on β with probability $(1+\eta)/2$, and a Cu atom on β with probability $(1-\eta)/2$. Averaging over the two cases gives $\alpha_1 = -\eta^2$ and $\alpha_2 = +\eta^2$ (dashed lines). Below T_c the Cowley parameters follow these curves: the long-range order explains almost all the local order. Above T_c, however, $\langle\eta^2\rangle$ is almost zero while α_1 stays clearly negative: at 800 K, $\langle\alpha_1\rangle = -0.20$ whereas $-\langle\eta^2\rangle = -0.05$, and at 1600 K $\langle\alpha_1\rangle$ is still -0.07. Cu atoms keep more Zn first neighbours than in a random alloy: this *short-range order* survives the loss of long-range order and fades slowly with temperature, as observed experimentally in ordering alloys [10].

4.2 Transition temperature

The energy, the order parameter and the two response functions all change abruptly around the same temperature: the alloy goes through a phase transition. Each quantity shows it in its own way:

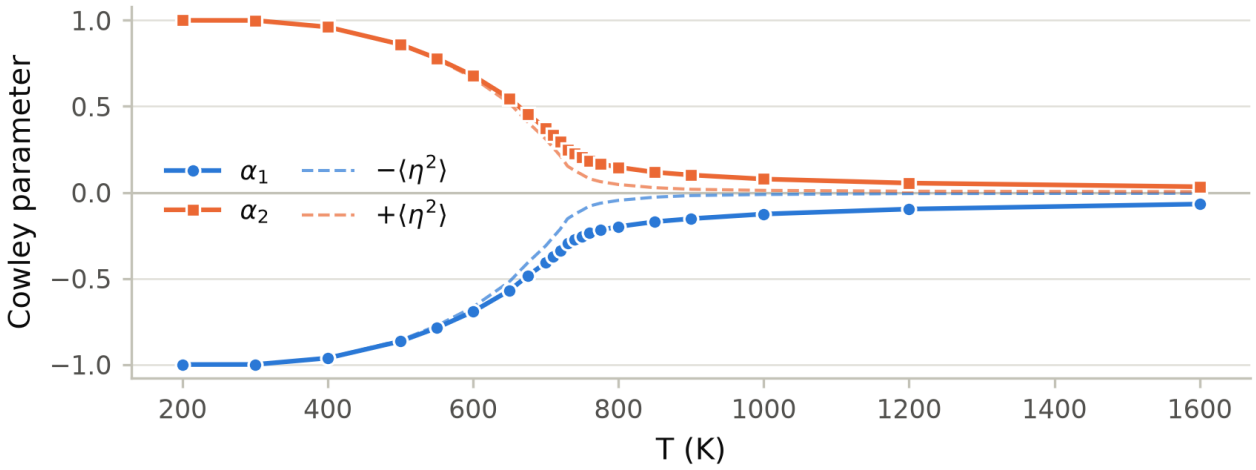


Figure 7: Cowley short-range order parameters for the first (α_1, circles) and second (α_2, squares) neighbour shells. The dashed lines $\mp\langle\eta^2\rangle$ are the values expected if the long-range order alone were present. The difference between symbols and dashed lines above 750 K is the short-range order.

- *Order parameter.* η distinguishes the two sublattices: it is non-zero in the ordered phase and zero in the disordered phase, where α and β become equivalent. Its fast drop is the loss of long-range order, which defines the transition. We take as an estimate of T_c the temperature of the steepest drop (minimum of $\mathrm{d}\langle|\eta|\rangle/\mathrm{d}T$, by finite differences).
- *Heat capacity.* $C_v = \mathrm{d}\langle E\rangle/\mathrm{d}T$ measures how fast the energy rises with temperature. At the transition, a small increase of T replaces many Cu–Zn first-neighbour pairs by Cu–Cu and Zn–Zn pairs, so the energy rises fast and C_v has a peak.
- *Susceptibility.* χ measures the fluctuations of η. Its peak shows that the order is the least stable at the transition, where the system wanders between ordered and disordered states (Figure 6). These wanderings are also seen in the energy: by Equation (4), the peak of C_v is a maximum of the energy fluctuations.

In an infinite crystal η would vanish exactly at T_c, where C_v and χ would be singular; in our small box the drop is rounded and the peaks are finite.

Table 3 gives the three estimates. They fall on the same grid temperature, 720 K. The grid step is 10 K there, but the peaks are flat at the top, between 700 and 730 K (Table 4): we retain $T_c \approx 720$ K.

Table 3: Transition temperature. The Monte Carlo values are read on the temperature grid of Table 2 (step 10 K between 700 and 760 K).

Method	T_c (K)
Maximum of the heat capacity C_v	720
Maximum of the susceptibility χ	720
Steepest drop of $\langle\|\eta\|\rangle$	720
Experiment (β-brass) [1–3]	$\approx$ 740–750

5 Discussion

Agreement with experiment. Our estimate of T_c is close to the experimental value, which is quite good for a model with only six parameters and no adjustment. This agreement should not be over-interpreted: the model ignores the vibrations of the atoms, the thermal expansion of the lattice and the difference in size between Cu and Zn. For comparison, a rigid-lattice Monte Carlo

study of CuZn with energies computed from first principles finds $T_c \approx 870\,\mathrm{K}$, about 120 K above experiment [3]: a good T_c is not a proof that every ingredient of the model is right.

Finite size and sampling. The position of the peaks depends on the size of the box: a study with several sizes (for example with the crossing of the Binder cumulant) would give T_c for an infinite crystal [7, 11]. Successive measurements are also correlated, especially near T_c where the order changes slowly (critical slowing down [12]). C_v and χ are the noisiest quantities, because they are variances of correlated series: with only 800 sweeps per temperature, a first version of our analysis gave 700, 720 and 730 K for the three estimates.

Algorithm. Exchanging only first neighbours conserves the composition and samples the correct equilibrium distribution, but it is slow to move order over long distances. Exchanges between any two sites would equilibrate faster at about the same cost per step.

6 Conclusion

We wrote and tested a Metropolis Monte Carlo code for the order–disorder transition of CuZn on a rigid bcc lattice. The tests on the perfect B2 structure and on random lattices give the expected values, and the energy updated move by move stays exact. The cohesive energy of the B2 phase is 2.484 eV/atom. Starting from B2, the system needs at most about 12 000 steps to reach equilibrium, so we discarded 50 000 steps before measuring during 4000 sweeps. The energy, the heat capacity, the long-range order parameter and its susceptibility all show the transition at $T_c \approx 720\,\mathrm{K}$, close to the experimental value (about 740 K). Above the transition, Cu and Zn still prefer to be first neighbours: short-range order survives the loss of long-range order. The main limits of this study are the small size of the box and the simple pair model; larger boxes and a finite-size analysis would be the next step.

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A Full results of the temperature scan

Table 4: Results of the temperature scan (conditions of Table 2). E: mean energy per atom (eV); C_v: heat capacity of the box (eV/K); χ: susceptibility (eV^{-1}).

T (K)	$\langle E\rangle/N$	C_v	$\langle\lvert\eta\rvert\rangle$	χ	$\langle\alpha_1\rangle$	$\langle\alpha_2\rangle$
200	−2.4840	0.0000	1.000	0	−1.000	1.000
300	−2.4840	0.0002	1.000	0	−1.000	1.000
400	−2.4825	0.0088	0.980	2	−0.962	0.961
500	−2.4788	0.0220	0.926	6	−0.865	0.861
550	−2.4757	0.0277	0.879	10	−0.787	0.779
600	−2.4720	0.0350	0.815	17	−0.692	0.678
650	−2.4673	0.0422	0.717	35	−0.570	0.547
675	−2.4639	0.0523	0.631	86	−0.485	0.455
700	−2.4610	0.0601	0.539	139	−0.409	0.373
710	−2.4596	0.0536	0.487	166	−0.374	0.335
720	−2.4582	0.0602	0.427	211	−0.337	0.295
730	−2.4565	0.0439	0.350	205	−0.296	0.249
740	−2.4556	0.0374	0.313	190	−0.272	0.224
750	−2.4549	0.0322	0.276	183	−0.254	0.205
760	−2.4542	0.0226	0.240	143	−0.235	0.185
775	−2.4535	0.0168	0.212	116	−0.218	0.167
800	−2.4528	0.0128	0.176	94	−0.199	0.146
850	−2.4517	0.0075	0.133	58	−0.170	0.117
900	−2.4510	0.0052	0.108	36	−0.152	0.101
1000	−2.4500	0.0032	0.088	22	−0.126	0.078
1200	−2.4489	0.0017	0.067	11	−0.097	0.054
1600	−2.4478	0.0008	0.056	5	−0.068	0.033

B Code and reproduction

The model is written in Python in `src/model.py` (one module, about 190 lines):

- the array `sous_reseau` holds the species of each site (0 = Cu, 1 = Zn); site i of cell (x, y, z) and sublattice s has index $i = 2(x + 6y + 36z) + s$, so even sites are α and odd sites β;
- `FIRST_NEIGHBOURS` and `SECOND_NEIGHBOURS` are the neighbour tables, built once at import;
- `get_full_energy()` computes Equation (1); `energy_after_flip(E, i, j)` returns the energy after exchanging i and j from the local bonds only, without changing the configuration; `flipping_proba` and `flipping_action` apply the Metropolis rule;
- `cowley()` and `order_parameter()` return (α_1, α_2) and η;
- `metropolis_before_eq(n, T)` returns the energy after each of n steps from B2 (equilibration test); `metropolis(n_eq, n_meas, T, measure_every)` returns the series of E, $|\eta|$, α_1, α_2 measured after equilibration.

The analysis notebook `analysis.ipynb` uses these functions interactively. All the figures and tables of this report are produced by `report/make_figures.py`, with the parameters of Table 2. From the project folder:

```
python -m venv .venv && .venv/bin/pip install -r requirements.txt
.venv/bin/python -B report/make_figures.py --rerun   # simulations, figures
cd report && latexmk -pdf report.tex slides.tex       # report and slides
```